

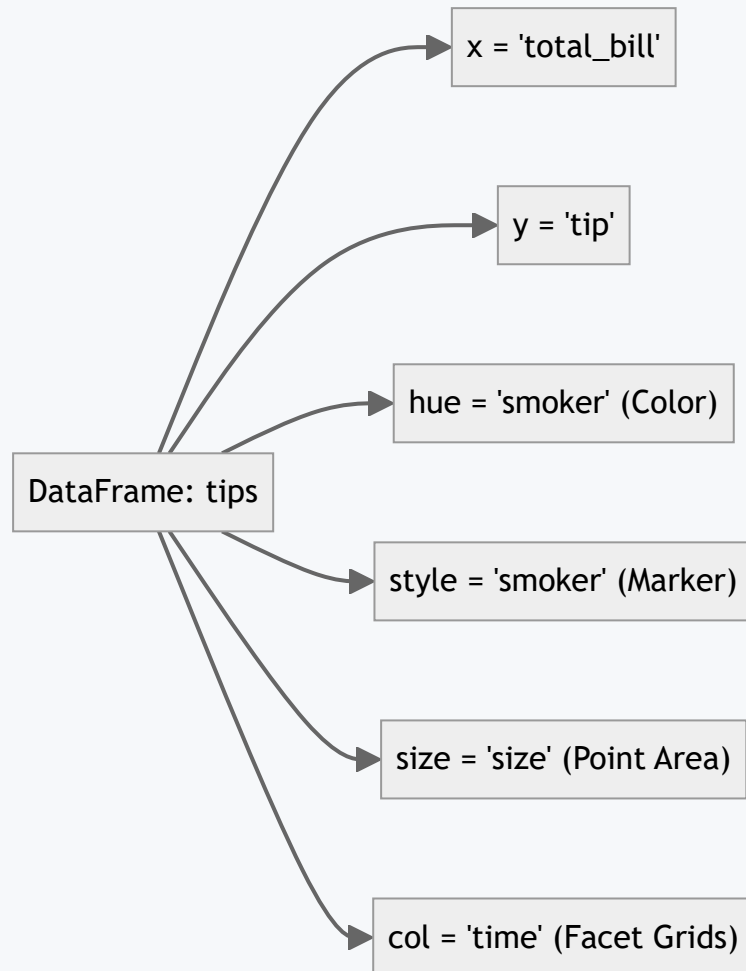
Day 19 - Lecture 19.11: Creating Plots with Seaborn (Semantic Mapping)

1. Overview & Semantic Dimensions

In standard plotting, each 2D chart represents 2 variables (X and Y). Seaborn leverages the **Grammar of Graphics** to allow up to **5 or 6 dimensions of data** on a single 2D plot using **semantic mappings**:

1. X -axis / Y -axis: Continuous quantitative positions.
 2. **hue**: Colors points/lines based on a categorical or continuous column.
 3. **style**: Changes marker symbols (circles, crosses, triangles) or line dash styles.
 4. **size**: Scales the area of markers or width of lines.
 5. **col** / **row**: Splits data across subplots (Faceting).
-

2. Multi-Dimensional Semantic Mapping with `sns.relplot()`



`sns.relplot()` Syntax:

```
sns.relplot(  
    data=tips,  
    x="total_bill",    # Quantitative X  
    y="tip",           # Quantitative Y  
    col="time",        # Creates 2 subplots (Lunch & Dinner)  
    hue="smoker",      # Color by smoker status (Yes / No)  
    style="smoker",     # Different marker by smoker status  
    size="size",       # Marker size proportional to party size (1 to 6 people)  
    palette="Set2"     # Color palette  
)
```

3. Code Implementation

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")

# Multi-dimensional relational plot
sns.set_theme(style="white")

g = sns.relplot(
    data=tips,
    x="total_bill",
    y="tip",
    col="time",
    hue="smoker",
    style="smoker",
    size="size",
    sizes=(30, 200),
    palette="Dark2",
    alpha=0.85
)

g.fig.subplots_adjust(top=0.85)
g.fig.suptitle("Multi-Dimensional Relationship: Bill vs Tip by Smoker, Party Size")
plt.show()

# Line mapping with relplot
x_vals = list(range(1, 11))
y_vals = [i**2 for i in x_vals]

sns.relplot(x=x_vals, y=y_vals, kind="line", marker="o", color="crimson")
plt.title("Quadratic Growth with relplot(kind='line')")
plt.show()
```

4. Key Takeaways

- **hue + style pairing:** When plotting scatter points, using `hue="smoker"`, `style="smoker"` ensures both color and shape change together, making the chart **accessible to colorblind viewers**.
- `sns.relplot()` is a **Figure-level function** that returns a `FacetGrid` object, allowing automated subplot generation via `col` and `row`.

